

CAPSTONE PROJECT

NOTEBOOK 04: MONGODB DATA INGESTION PIPELINE

Purpose:

This notebook creates a MongoDB pipeline for the FMA dataset.

It loads metadata and features, prepares unified track records,

and inserts them into MongoDB.

In [1]:

```
# =====  
# 1. IMPORT LIBRARIES  
# =====  
  
import pandas as pd  
import numpy as np  
import os  
from pymongo import MongoClient
```

In [2]:

```
# =====
# 2. SET PATHS
# =====

metadata_path = "../data/raw/metadata"

tracks_path = os.path.join(metadata_path, "tracks.csv")
features_path = os.path.join(metadata_path, "features.csv")
genres_path = os.path.join(metadata_path, "genres.csv")
```

In [3]:

```
# =====
# 3. LOAD DATA
# =====

tracks = pd.read_csv(
    tracks_path,
    header=[0, 1],
    index_col=0
)

features = pd.read_csv(
    features_path,
    header=[0, 1, 2],
    index_col=0
)

genres = pd.read_csv(genres_path)

print("Tracks:", tracks.shape)
print("Features:", features.shape)
print("Genres:", genres.shape)
```

Tracks: (106574, 52)
 Features: (106574, 518)
 Genres: (163, 5)

In [4]:

```
# =====
# 4. FILTER TO SMALL SUBSET FIRST
# =====
# We start with small because it is safer for testing.
# Later, this same pipeline can be used for medium or large.

subset_name = "small"

tracks_subset = tracks[tracks[("set", "subset")] == subset_name]
features_subset = features.loc[tracks_subset.index]

print("Subset:", subset_name)
```

```
print("Tracks subset:", tracks_subset.shape)
print("Features subset:", features_subset.shape)
```

Subset: small
 Tracks subset: (8000, 52)
 Features subset: (8000, 518)

In [5]:

```
# =====
# 5. KEEP ONLY TRACKS WITH GENRE LABELS
# =====

valid = tracks_subset[("track", "genre_top")].notna()

tracks_subset = tracks_subset[valid]
features_subset = features_subset[valid]

print("After removing missing labels:")
print("Tracks:", tracks_subset.shape)
print("Features:", features_subset.shape)
```

After removing missing labels:
 Tracks: (8000, 52)
 Features: (8000, 518)

In [6]:

```
# =====
# 6. CLEAN FEATURE COLUMNS
# =====
# MongoDB does not work well with complex multi-index column names.
# We flatten feature columns into readable names.

features_subset = features_subset.select_dtypes(include=["number"])
features_subset = features_subset.replace([np.inf, -np.inf], np.nan)
features_subset = features_subset.fillna(features_subset.mean())

features_subset.columns = [
    "_".join([str(part) for part in col if str(part) != "nan"])
    for col in features_subset.columns
]

print("Cleaned features:", features_subset.shape)
```

Cleaned features: (8000, 518)

In [7]:

```
# =====
# 7. FUNCTION TO CREATE AUDIO PATH
# =====
# FMA audio files are stored using the track ID.
# Example:
```

```
# track_id = 2
# formatted = 000002
# folder = 000
# path = ../data/raw/audio/fma_small/000/000002.mp3

def create_audio_path(track_id, audio_base_path="../data/raw/audio/fma_small"):
    track_id_str = f"{int(track_id):06d}"
    folder = track_id_str[:3]
    return os.path.join(audio_base_path, folder, track_id_str + ".mp3")
```

In [8]:

```
# =====
# 8. CREATE MONGODB RECORDS
# =====
# Each record represents one track.
# This combines:
# - metadata from tracks.csv
# - split/subset information
# - genre label
# - audio path
# - selected numeric features

records = []

for track_id in tracks_subset.index:

    track_row = tracks_subset.loc[track_id]
    feature_row = features_subset.loc[track_id]

    record = {
        "track_id": int(track_id),
        "subset": str(track_row[("set", "subset")]),
        "split": str(track_row[("set", "split")]),

        "genre": {
            "genre_top": str(track_row[("track", "genre_top")]),
            "genres": str(track_row[("track", "genres")]),
            "genres_all": str(track_row[("track", "genres_all")])
        },

        "track_metadata": {
            "title": str(track_row[("track", "title")]),
            "duration": float(track_row[("track", "duration")]),
            "listens": int(track_row[("track", "listens")]),
            "interest": int(track_row[("track", "interest")]),
            "language_code": str(track_row[("track", "language_code")]),
            "license": str(track_row[("track", "license")])
        },

        "artist_metadata": {
            "artist_id": int(track_row[("artist", "id")]),
            "artist_name": str(track_row[("artist", "name")]),
            "artist_location": str(track_row[("artist", "location")])
        }
    }
```

```
        "album_metadata": {
            "album_id": int(track_row[("album", "id")]),
            "album_title": str(track_row[("album", "title")]),
            "album_type": str(track_row[("album", "type")])
        },

        "audio_path": create_audio_path(track_id),

        "features": feature_row.to_dict()
    }

    records.append(record)

print("Total records prepared:", len(records))
print("Example record:")
records[0]
```

Total records prepared: 8000

Example record:

```
Out[8]: {'track_id': 2,
'subset': 'small',
'split': 'training',
'genre': {'genre_top': 'Hip-Hop', 'genres': '[21]', 'genres_all': '[21]'},
'track_metadata': {'title': 'Food',
'duration': 168.0,
'listens': 1293,
'interest': 4656,
'language_code': 'en',
'license': 'Attribution-NonCommercial-ShareAlike 3.0 International'},
'artist_metadata': {'artist_id': 1,
'artist_name': 'AWOL',
'artist_location': 'New Jersey'},
'album_metadata': {'album_id': 1,
'album_title': 'AWOL - A Way Of Life',
'album_type': 'Album'},
'audio_path': '../data/raw/audio/fma_small\\000\\000002.mp3',
'features': {'chroma_cens_kurtosis_01': 7.1806526184,
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```

```
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```

```
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```
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```

```
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```

```
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```

In [9]:

```
# =====
# 9. CONNECT TO MONGODB
# =====
# This assumes MongoDB is running locally.
# If using MongoDB Atlas, replace the connection string.

client = MongoClient("mongodb://localhost:27017/")

db = client["fma_capstone"]
collection = db["tracks_small"]

print("Connected to MongoDB successfully.")
```

Connected to MongoDB successfully.

In [10]:

```
# =====
# 10. INSERT RECORDS INTO MONGODB
# =====
# We delete old records first so the notebook can be rerun safely.

collection.delete_many({})

collection.insert_many(records)

print("Inserted records:", collection.count_documents({}))
```

Inserted records: 8000

In [11]:

```
# =====
# 11. TEST BASIC QUERIES
# =====

print("Total records:", collection.count_documents({}))

print("Training records:", collection.count_documents({"split": "training"}))
print("Validation records:", collection.count_documents({"split": "validation"}))
print("Test records:", collection.count_documents({"split": "test"}))
```

Total records: 8000

Training records: 6400

Validation records: 800

Test records: 800

In [12]:

```
# =====
# 12. COUNT TRACKS BY GENRE
# =====

genre_counts = collection.aggregate([
    {
        "$group": {
            "_id": "$genre.genre_top",
            "count": {"$sum": 1}
        }
    },
    {
        "$sort": {"count": -1}
    }
])

for item in genre_counts:
    print(item)
```

```
{'_id': 'Instrumental', 'count': 1000}
{'_id': 'Experimental', 'count': 1000}
{'_id': 'Pop', 'count': 1000}
{'_id': 'Hip-Hop', 'count': 1000}
{'_id': 'Folk', 'count': 1000}
{'_id': 'Electronic', 'count': 1000}
{'_id': 'Rock', 'count': 1000}
{'_id': 'International', 'count': 1000}
```

In [13]:

```
# =====
# 13. VIEW ONE SAMPLE RECORD
# =====
```

```
sample = collection.find_one()
```

```
sample
```

```

Out[13]: {'_id': ObjectId('69f2f1163e09dd500b484235'),
  'track_id': 2,
  'subset': 'small',
  'split': 'training',
  'genre': {'genre_top': 'Hip-Hop', 'genres': '[21]', 'genres_all': '[21]'},
  'track_metadata': {'title': 'Food',
    'duration': 168.0,
    'listens': 1293,
    'interest': 4656,
    'language_code': 'en',
    'license': 'Attribution-NonCommercial-ShareAlike 3.0 International'},
  'artist_metadata': {'artist_id': 1,
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```

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